MTH 202 Chapter 5: Topics In Vector Calculus

5.1 Vector Fields

Up to this point in MTH 202 we have studied functions of the form $\mathbf{r}:\mathbb{R}\to\mathbb{R}^n$ and $f:\mathbb{R}^n\to\mathbb{R}$. In this final chapter of MTH 202 we turn our focus to the most general situation. Namely we will study functions of the form $\mathbf{F}:\mathbb{R}^m\to\mathbb{R}^n$.

Definition:

1. Let D be a set in $\mathbb{R}^2$. A vector field on $\mathbb{R}^2$ is a function $\mathbf{F}$ that assigns to each point (x,y) in D a two-dimensional vector $\mathbf{F}(x,y)$.
2. Let E be a set in $\mathbb{R}^3$. A vector field on $\mathbb{R}^3$ is a function $\mathbf{F}$ that assigns to each point (x,y,z) in E a three-dimensional vector $\mathbf{F}(x,y,z)$.

Example 1: Use a picture to illustrate the vector field $\mathbf{F}(x,y)=\langle -y,x\rangle$.

Remark: Vector fields have a wide variety of applications. For example, a vector field on $\mathbb{R}^2$ can be used to model the amount of force exerted by an electric charge on some charge located at the point (x, y) on a two dimensional plate.

A vector field on $\mathbb{R}^3$ can be used to model the force of the wind at some point (x, y, z) in space.

A vector field on $\mathbb{R}^3$ can be used to model the current of the ocean at some point (x, y, z) in the ocean.

A vector field on $\mathbb{R}^3$ can be used to model the force due to gravity at some point (x, y, z) in space.

A vector field in $\mathbb{R}^3$ can be used to model the velocity of a moving fluid at some point (x, y, z) in space.

<u>**Example 2:**</u> The gravitational force from an object of mass M centered at the origin acting on a small object of mass m at $\mathbf{x} = \langle x, y, z\rangle$ is $\mathbf{F}(x, y, z) = -\frac{mMG}{\|\mathbf{x}\|^3}\mathbf{x}$ where G is the universal gravitational constant. Suppose a large three dimensional object has center of mass at the origin and has mass equal to 100,000 kg. Suppose a small object of mass 5 kg is located at the terminal point of the position vector $\mathbf{x} = \langle 10, 5, 10\rangle$ (you may assume distance is in meters). What is the gravitational force acting on the small object (due to the large object)?

Definition: If f is a scalar function of two or three variables, $\vec{\nabla} f$ is called the <u>gradient vector field of f</u>.

Definition: A vector field $\mathbf{F}$ is called a <u>conservative vector field</u> if it is the gradient of some scalar function f. Mathematically, this means there must exist a scalar function f so that $\mathbf{F} = \vec{\nabla} f$. In this situation f is called a <u>potential function</u> for $\mathbf{F}$.

Example 3: Prove that $\mathbf{F}(x,y)=\langle x,y\rangle$ is a conservative vector field.

Question: Is $f(x,y)=\frac{1}{2}x^2+\frac{1}{2}y^2$ *the only scalar function that would have worked in the above proof?*

Note: For now, the only technique we have for proving a vector field is conservative is to guess and check until we successfully find a potential function for the vector field. We will develop more sophisticated techniques as in section 5.3.

Note: Right now, we do not have ANY technique for proving that a vector field is not conservative. We will develop techniques for proving a vector field is not conservative in section 5.3.

Example 4: Consider $f(x,y,z)=xye^z$.

a) Find the gradient vector field of f.

b) Is the vector field you found in (a) conservative? Explain.

Note: Later in chapter 5 we will discover that conservative vector fields have some very nice properties.